

From Predictive Agreement to Geometric Agreement: Stability of Fisher Connections and Semantic Transport in Neural Representations

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Abstract

A probabilistic decoder equips its predictively identifiable hidden-state quotient with the pullback Fisher metric and Amari–Chentsov tensor. We establish a stability hierarchy for comparing this geometry across models. Exact equality of predictive images preserves every Amari α -connection, transport, curvature, and intrinsic predictive field. Uniform KL convergence alone does not imply even pointwise Fisher-metric convergence. General α -transport is stable under uniform interiority, nondegeneracy, and C^2 predictive agreement, but the resulting worst-case constants are poorly adapted to large vocabularies. For the Levi–Civita connection we remove both the vocabulary-size and minimum-token-probability dependence: C^2 closeness of the square-root predictive maps controls the connection and transport, and integrated KL plus a uniform Hilbert-valued Sobolev bound supplies this closeness at an explicit rate. A paired boundary Bernoulli construction proves that the same square-root regularity cannot control any fixed nonzero- α connection without additional information. Score-moment identities identify the required raised third-order moment and yield probability-floor-free formulas at affine softmax heads. Finally, conditional Fisher variance is the exact obstruction to descending a semantic effect through a noninjective predictive map; it vanishes identically when that map is injective. Constant vector reuse at an affine head is exponential transport on the ambient intervention family, so superiority of any other connection is a held-out empirical hypothesis rather than a consequence of pretraining.

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1 Introduction

Modern neural networks compute with vectors while producing probability distributions. For an autoregressive language model, a hidden state h entering the output head determines

$$F(h) = p(\cdot \mid h) \in \Delta_N^\circ, \tag{1}$$

where Δ_N° is the interior of the categorical simplex over a vocabulary of size N . The hidden coordinate remains a point of $\mathbb{R}^d$. Nevertheless, its predictively visible component parameterizes a statistical model and inherits the Fisher geometry of the decoder.

Exact equality of identifiable predictive images gives an isomorphism of their induced statistical structures. Approximate equality at the loss level is weaker: cross-entropy controls probability

values on the data distribution, whereas Fisher metrics and affine connections depend on derivatives of the predictive map. Geometric equivalence also does not imply that a semantic operation is determined by the next-token distribution, because hidden states can contain decoder-null or semantically relevant fiber directions.

At a linear softmax head, constant hidden-vector reuse is exactly the affine operation selected by the flat exponential connection. Ordinary addition is therefore one connection-dependent rule rather than an alternative to geometry. The resulting question is which connection, if any, makes a specified semantic effect transferable across contexts and models.

The analysis proceeds through six results:

1. Exact predictive naturality: two identifiable parameterizations of the same predictive family preserve the Fisher metric, Amari-Chentsov tensor, every α -connection, parallel transport, and curvature up to reparameterization; every common smooth field on the shared predictive image pulls back equivariantly.
2. Separation of predictive and geometric convergence: a smooth, uniformly regular Bernoulli family converges uniformly in KL while its Fisher metric fails to converge at a fixed point.
3. Quantitative C^k stability: on compact uniformly interior and uniformly identifiable families, C^2 predictive closeness controls the defect between aligned α -connections and the commutator of their parallel transports, while C^3 closeness controls curvature.
4. A sharp connection-selective boundary result: square-root regularity removes explicit vocabulary-size and minimum-probability factors from Levi-Civita bounds, while paired Bernoulli families converge in square-root C^2 even as every fixed nonzero- α connection defect diverges.
5. Loss-to-geometry implications under explicit regularity: integrated KL plus uniform Sobolev control of either probability coordinates or square-root coordinates yields transport convergence; the latter result is vocabulary independent for Levi-Civita transport.
6. Semantic conditional-variance decompositions: two orthogonal identities separately isolate failure to descend through a noninjective predictive map from connection mismatch and from cross-model conditional-mean-field mismatch.

These results define a concrete comparison protocol: choose a predictive quotient and a cross-model alignment, estimate infinitesimal semantic effects on a declared continuous chart, and compare exponential, Levi-Civita, mixture, and learned- α transport through commutation error and held-out behavioral transfer. The square-root sphere representation itself is classical; the contribution here is the quantitative implication-separation package and its decoder interpretation.

2 Related work

Information geometry equips regular statistical models with the Fisher metric, the Amari-Chentsov cubic tensor, and dual affine connections [2, 3, 6]. Invariance under sufficient statistics and the uniqueness of the Fisher and Amari-Chentsov tensors are classical [5, 8]. The square-root representation identifies the categorical Fisher metric with the metric induced from a Euclidean sphere. The distinction between affine flatness and Levi-Civita flatness is standard; global exponential or mixture coordinates do not force zero Riemannian curvature [7].

Pullback geometry in learned latent spaces is also established. Arvanitidis et al. [4] pull Fisher-Rao geometry through probabilistic decoders. Zhang [20] explicitly connects a language model’s

hidden-to-predictive map, likelihood training, and pullback geometry. For softmax representations, Park et al. [13] develop an information-geometric account of probing and steering, while Wang and Zhao [17] pull the output Fisher metric through transformer layers and show large deviations from ambient Euclidean geometry. Sarfati et al. [15] empirically recover curved representation manifolds for controlled posterior beliefs and use field-aware interventions to remain on the belief family. Mabrok [12] instead proposes a Fisher-metric latent semantic manifold partitioned into token regions and studies distortion induced by vocabulary discretization. These results overlap the decoder/manifold motivation and the case for geometry-aware interventions, but they do not supply the predictive-agreement stability implications proved below.

Cross-model representation convergence has been studied from several directions. The Platonian Representation Hypothesis proposes convergence toward shared statistical structure [11]. Yu et al. [19] align distinct neural latent geometries with pullback metrics and relative geodesic representations. Hu et al. [10] model contextual transformations as vector fields and report shared displacement structure across language-model families. Path dependence and holonomy have also been proposed as representation diagnostics [16]. The stability analysis below asks when predictive agreement forces agreement of connection-dependent transport.

Relative to all of these, the contribution here is the quantitative stability and separation package: square-root C^2 control of Levi-Civita transport with constants carrying no explicit vocabulary-size or token-probability factors, a paired Bernoulli construction showing that no fixed nonzero- α connection admits the same control without third-score-moment information, an integrated-KL-plus-Sobolev route from training risk to transport agreement, and the conditional-variance obstruction to semantic descent. To our knowledge, none of the works above states these stability or separation theorems; they supply the geometric setting and the empirical motivation rather than the agreement theory.

Continuity of Fisher information requires control of both distributions and their derivatives; Rezakhani et al. [14] prove such bounds in classical and quantum settings. Quantitative relaxations of sufficient statistics based on bi-Lipschitz Fisher metrics are studied by Yamaguchi and Nozawa [18]. Here a fixed comparison diffeomorphism aligns the predictive maps; their closeness is then propagated through every Amari α -connection to an explicit parallel-transport commutator bound. Sobolev regularity supplies the derivative control needed to derive this closeness from KL convergence.

3 Predictive statistical manifolds

3.1 The categorical simplex

Let

$$\Delta_N^\circ = \left\{ p \in \mathbb{R}^N : p_a > 0, \sum_{a=1}^N p_a = 1 \right\}. \quad (2)$$

For tangent vectors u, v, w satisfying $\sum_a u_a = \sum_a v_a = \sum_a w_a = 0$, define

$$g_p^\Delta(u, v) = \sum_{a=1}^N \frac{u_a v_a}{p_a}, \quad (3)$$

$$C_p^\Delta(u, v, w) = \sum_{a=1}^N \frac{u_a v_a w_a}{p_a^2}. \quad (4)$$

These are the Fisher metric and Amari-Chentsov tensor. A smooth predictive map

$$F : M \longrightarrow \Delta_N^\circ \quad (5)$$

induces

$$g_F = F^*g^\Delta, \quad C_F = F^*C^\Delta. \quad (6)$$

When F is an immersion, g_F is a Riemannian metric (Lemma B.8). In local coordinates $x = (x^1, \dots, x^m)$, writing $p_a = F_a(x)$,

$$(g_F)_{ij} = \sum_a \frac{\partial_i p_a \partial_j p_a}{p_a}, \quad (7)$$

$$(C_F)_{ijk} = \sum_a \frac{\partial_i p_a \partial_j p_a \partial_k p_a}{p_a^2}. \quad (8)$$

Raise the final index of C_F using g_F and define K_F by

$$g_F(K_F(X, Y), Z) = C_F(X, Y, Z). \quad (9)$$

We use the convention

$$\nabla_F^{(\alpha)} = \nabla_F^{\text{LC}} - \frac{\alpha}{2} K_F. \quad (10)$$

The case $\alpha = 0$ is Levi-Civita by definition. That $\alpha = 1$ and $\alpha = -1$ are the exponential and mixture connections on a regular exponential family is not a consequence of (10) alone; it is proved in Lemma B.3.

3.2 Predictive quotients

Neural predictive maps are commonly nonidentifiable. Define

$$h \sim_F h' \iff F(h) = F(h'). \quad (11)$$

When the fibers are regular, the identifiable state space is the quotient $M_F = M/\sim_F$; Remark B.7 states the hypothesis under which this quotient is a smooth manifold and notes that none of the estimates below require it. All geometric statements below concern this quotient or a chosen horizontal subbundle on which the pullback metric is nondegenerate.

For a linear softmax decoder,

$$F(h) = \text{softmax}(Wh + b), \quad (12)$$

the null space is

$$\mathcal{N} = \{u : Wu \in \text{span}\{\mathbf{1}\}\}. \quad (13)$$

The quotient $\mathbb{R}^d/\mathcal{N}$ embeds smoothly into the simplex (Lemma B.6; an injective immersion need not be an embedding, so this requires proof), and before quotienting

$$G(h) = W^\top (\text{diag } p_h - p_h p_h^\top) W, \quad \ker G(h) = \mathcal{N}. \quad (14)$$

The kernel identification is Lemma B.6(i). This is the rigorous sense in which the predictively identifiable component of a hidden space is a statistical manifold. A finite set of naturally occurring hidden states remains a point cloud unless a smooth intervention chart or interpolation model is supplied.

3.3 Ordinary addition selects a connection

For the affine softmax family

$$p_h(a) = \exp(w_a^\top h + b_a - A(h)), \quad (15)$$

natural hidden coordinates are exponential-affine, so constant coordinate vectors are $\nabla^{(1)}$ -parallel by Lemma B.3. More explicitly,

$$p_{h_r+h_q-h_p}(a) = \frac{p_{h_r}(a)p_{h_q}(a)/p_{h_p}(a)}{\sum_b p_{h_r}(b)p_{h_q}(b)/p_{h_p}(b)}. \quad (16)$$

Thus hidden vector arithmetic in the ambient affine-head intervention family is exactly exponential transport, not the absence of geometry. Levi-Civita transport instead preserves Fisher lengths and angles, while mixture transport preserves the dual affine structure. Naturally occurring post-normalization states can occupy a lower-dimensional subset that is not closed under this arithmetic. Which connection best models a semantic operation on a declared intervention chart is an empirical question.

4 Exact predictive naturality

Exact predictive equivalence determines the corresponding metric, tensors, connections, and transports.

Theorem 4.1 (Exact predictive naturality). *Let $F_A : M_A \rightarrow \Delta_N^\circ$ and $F_B : M_B \rightarrow \Delta_N^\circ$ be smooth embeddings onto the same embedded predictive image P . Define*

$$\Phi = F_B^{-1} \circ F_A : M_A \longrightarrow M_B. \quad (17)$$

Then

$$\Phi^* g_B = g_A, \quad \Phi^* C_B = C_A. \quad (18)$$

For every $\alpha \in \mathbb{R}$,

$$d\Phi(\nabla_A^{(\alpha)} X Y) = \nabla_B^{(\alpha)} d\Phi X (d\Phi Y). \quad (19)$$

For every piecewise C^1 path γ ,

$$d\Phi_{\gamma(1)} P_\gamma^{A,\alpha} = P_{\Phi \circ \gamma}^{B,\alpha} d\Phi_{\gamma(0)}. \quad (20)$$

Curvature tensors intertwine, and holonomies of corresponding loops are conjugate.

Proof. Since $F_A = F_B \circ \Phi$, functoriality of pullback gives

$$\Phi^* g_B = \Phi^* F_B^* g^\Delta = (F_B \circ \Phi)^* g^\Delta = F_A^* g^\Delta = g_A, \quad (21)$$

and identically for C . A Riemannian isometry preserves the Levi-Civita connection. Preservation of g and C implies preservation of K , so (10) gives (19). Applying $d\Phi$ to an A -parallel field and using uniqueness of the linear transport ODE gives (20). Curvature naturality follows from its definition as the commutator of covariant derivatives, and loop holonomies are consequently conjugate. $\square$

Corollary 4.2 (Intrinsic semantic fields). *Let $S_t : P \rightarrow P$ be a smooth semantic flow with infinitesimal field*

$$Z_P(p) = \left. \frac{d}{dt} \right|_{t=0} S_t(p). \quad (22)$$

Define model fields X_i by $dF_i(X_i) = Z_P \circ F_i$. Then

$$d\Phi(X_A) = X_B \circ \Phi. \quad (23)$$

Proof. Differentiate $F_A = F_B \circ \Phi$ and use the definitions of the two lifted fields:

$$\begin{aligned} dF_B(d\Phi(X_A)) &= dF_A(X_A) = Z_P \circ F_A \\ &= Z_P \circ F_B \circ \Phi = dF_B(X_B \circ \Phi). \end{aligned} \quad (24)$$

Because F_B is an embedding, dF_B is injective at every point. Hence $d\Phi(X_A) = X_B \circ \Phi$. $\square$

The condition is substantive: a context operation need not descend to P . If two hidden states have the same prediction before an intervention but distinct predictive effects afterward, then no intrinsic field Z_P exists on the predictive quotient. Section 8 quantifies this obstruction.

5 Prediction convergence does not imply geometric convergence

Let $P(\cdot | c)$ be the data conditional and $q(\cdot | c)$ a model. Population cross-entropy decomposes as

$$\mathcal{L}(q) - H_P(Y | C) = \mathbb{E}_C D_{\text{KL}}(P(\cdot | C) \| q(\cdot | C)). \quad (25)$$

The logarithmic score is strictly proper [9]. If models q_A, q_B have excess risks ϵ_A, ϵ_B , then Pinsker's inequality, the triangle inequality, and Jensen's inequality for $\sqrt{\cdot}$ yield (Lemma B.5)

$$\mathbb{E}_C \|q_A(\cdot | C) - q_B(\cdot | C)\|_1 \leq \sqrt{2\epsilon_A} + \sqrt{2\epsilon_B}. \quad (26)$$

This makes output agreement natural under realizability and global population-risk optimization. It provides no derivative control.

Proposition 5.1 (Regular KL-to-geometry counterexample). *There exist smooth, uniformly interior, regular Bernoulli predictive families p_n, p_* on $[-1, 1]$ such that*

$$\sup_x D_{\text{KL}}(p_*(x) \| p_n(x)) = O(n^{-2}), \quad (27)$$

but $g_{p_n}(0)$ does not converge to $g_{p_}(0)$.*

Proof. Let $p_*(x) = \text{Ber}(r(x))$ and $p_n(x) = \text{Ber}(r_n(x))$, where

$$r(x) = \frac{1}{3} + \frac{x}{20}, \quad r_n(x) = r(x) + \frac{1}{100n} \sin(nx). \quad (28)$$

All probabilities remain in a fixed compact subset of $(0, 1)$. Moreover,

$$r'_n(x) = \frac{1}{20} + \frac{1}{100} \cos(nx) \in [0.04, 0.06], \quad (29)$$

so every map is a regular immersion with a uniform rank bound. Uniform interiority and Taylor's theorem give $\sup_x D_{\text{KL}}(p_* \| p_n) = O(n^{-2})$. The Bernoulli pullback metric is

$$g(x) = \frac{r'(x)^2}{r(x)(1-r(x))} dx^2. \quad (30)$$

At $x = 0$, $r_n(0) = r(0) = 1/3$, but $r'_n(0) = 3/50$ and $r'(0) = 1/20$. Hence

$$g_n(0) = \frac{81}{5000} dx^2 \neq \frac{9}{800} dx^2 = g_*(0). \quad (31)$$

$\square$

This example rules out an unconditional theorem of the form “cross-entropy convergence implies Fisher convergence.” The oscillations have bounded first derivative but unbounded higher regularity, anticipating the Sobolev condition in Section 7.

There is a second nonidentifiability. For any hidden diffeomorphism ψ ,

$$H' = \psi \circ H, \quad F' = F \circ \psi^{-1} \quad (32)$$

produces exactly the same predictions while changing ambient Euclidean coordinates. Pullback Fisher geometry handles this gauge covariantly, but loss alone cannot identify a privileged hidden Euclidean metric.

6 Stability of Fisher geometry and transport

We now quantify how derivative-level predictive agreement propagates to geometry. We state the result on a common Euclidean chart. For two latent manifolds related by a comparison diffeomorphism Φ , replace the second map by $\tilde{p} = F_B \circ \Phi$.

Let $K \subseteq U \subset \mathbb{R}^m$ and let $p, \tilde{p} : U \rightarrow \Delta_N^\circ$. For an integer k , use the componentwise norm

$$\|p\|_{C^k(K)} = \max_{a, |\beta| \leq k} \sup_{x \in K} |\partial^\beta p_a(x)|. \quad (33)$$

Assumption 6.1 (Uniform regularity). For constants $\eta, \lambda, B_2 > 0$,

$$p_a(x), \tilde{p}_a(x) \geq \eta, \quad (34)$$

$$\|p\|_{C^2(K)} + \|\tilde{p}\|_{C^2(K)} \leq B_2, \quad (35)$$

$$\lambda_{\min}(g_p(x)), \lambda_{\min}(g_{\tilde{p}}(x)) \geq \lambda \quad (36)$$

for every a and $x \in K$.

The probability floor excludes the singular simplex boundary. The spectral floor excludes decoder-null directions after quotienting and prevents inverse-metric instability.

Theorem 6.2 (C^2 stability of α -transport). *Under Assumption 6.1, fix $A > 0$. There is a constant*

$$C_\Gamma = C_\Gamma(m, N, A, \eta^{-1}, \lambda^{-1}, B_2) \quad (37)$$

such that, uniformly for $|\alpha| \leq A$,

$$\begin{aligned} & \|g_p - g_{\tilde{p}}\|_{C^1(K)} + \|C_p - C_{\tilde{p}}\|_{C^0(K)} \\ & + \|\Gamma_p^{(\alpha)} - \Gamma_{\tilde{p}}^{(\alpha)}\|_{C^0(K)} \leq C_\Gamma \|p - \tilde{p}\|_{C^2(K)}. \end{aligned} \quad (38)$$

Let $\gamma : [0, 1] \rightarrow K$ be piecewise C^1 , with Euclidean length L_γ . If

$$M_\Gamma = \max\{\|\Gamma_p^{(\alpha)}\|_{C^0}, \|\Gamma_{\tilde{p}}^{(\alpha)}\|_{C^0}\}, \quad (39)$$

where the norm is the induced bilinear operator norm, then

$$\|P_\gamma^{p, \alpha} - P_\gamma^{\tilde{p}, \alpha}\|_{\text{op}} \leq L_\gamma e^{M_\Gamma L_\gamma} C_\Gamma \|p - \tilde{p}\|_{C^2(K)}. \quad (40)$$

Proof. Equations (7) and (8) are finite sums of rational functions of p and its first derivatives. Their denominators are controlled by η . Differentiating (7) once uses at most second derivatives, so the first two terms of (38) follow from the mean-value inequality on the compact *convex* regularity class; Lemma B.9 carries this out and records why convexity is needed.

Lemma B.1 gives the lower-index α -coefficients directly in probability coordinates:

$$\Gamma_{ij,k}^{(\alpha)} = \sum_a \left[\frac{\partial_{ij} p_a \partial_k p_a}{p_a} - \frac{1 + \alpha}{2} \frac{\partial_i p_a \partial_j p_a \partial_k p_a}{p_a^2} \right]. \quad (41)$$

Thus their difference is $O(\|p - \tilde{p}\|_{C^2})$. Raising the final index requires inverse metrics. The identity

$$g_p^{-1} - g_{\tilde{p}}^{-1} = g_p^{-1} (g_{\tilde{p}} - g_p) g_{\tilde{p}}^{-1} \quad (42)$$

and the spectral floor λ give inverse-metric stability, proving (38).

Along γ , transport solves the linear ODE

$$\dot{V}(t) = -\Gamma^{(\alpha)}(\gamma(t))[\dot{\gamma}(t)]V(t). \quad (43)$$

Duhamel's formula bounds the difference of the two fundamental solutions by the integral of the coefficient defect multiplied by their growth factors. Gronwall's inequality then yields (40); the two growth factors multiply to a constant along the path, which is what produces the single exponential rather than $e^{2M_\Gamma L_\gamma}$ (Lemma B.10). $\square$

Corollary 6.3 (Cross-model transport commutator). *Let $F_A : M_A \rightarrow \Delta_N^\circ$, $F_B : M_B \rightarrow \Delta_N^\circ$, and let $\Phi : M_A \rightarrow M_B$ be a C^2 local diffeomorphism. Suppose F_A and $F_B \circ \Phi$ satisfy Assumption 6.1 and*

$$\|F_A - F_B \circ \Phi\|_{C^2(K)} \leq \varepsilon. \quad (44)$$

Then

$$\begin{aligned} \|\mathrm{d}\Phi_{\gamma(1)} P_\gamma^{A,\alpha} - P_{\Phi \circ \gamma}^{B,\alpha} \mathrm{d}\Phi_{\gamma(0)}\|_{\mathrm{op}} \\ \leq L_\gamma e^{M_\Gamma L_\gamma} C_\Gamma \varepsilon, \end{aligned} \quad (45)$$

where the operator norm measures inputs with the background norm on $T_{\gamma(0)} M_A$ and outputs with the aligned tangent norm on $T_{\Phi(\gamma(1))} M_B$, defined as the pullback of the background norm through the alignment:

$$\|w\|_{\mathrm{al},x} := \|\mathrm{d}\Phi_x^{-1} w\|, \quad w \in T_{\Phi(x)} M_B. \quad (46)$$

In these norms the left-hand side equals $\|P_\gamma^{A,\alpha} - \mathrm{d}\Phi_{\gamma(1)} P_{\Phi \circ \gamma}^{B,\alpha} \mathrm{d}\Phi_{\gamma(0)}\|_{\mathrm{op}}$ on M_A , which is the quantity the proof bounds. If an independently chosen background norm is used on TM_B instead, the right-hand side acquires the additional factor $\sup_{x \in K} \|\mathrm{d}\Phi_x\|_{\mathrm{op}}$, which must then be stated. At $\varepsilon = 0$, this reduces to Theorem 4.1.

Proof. The connection induced by $F_B \circ \Phi$ is the pullback of the connection induced by F_B . Its transport is conjugated by $\mathrm{d}\Phi$. Apply Theorem 6.2 on M_A . $\square$

Theorem 6.4 (C^3 curvature stability). *Assume the hypotheses of Theorem 6.2, replace the C^2 bound by*

$$\|p\|_{C^3(K)} + \|\tilde{p}\|_{C^3(K)} \leq B_3, \quad (47)$$

and retain the probability and spectral floors. Then

$$\|R_p^{(\alpha)} - R_{\tilde{p}}^{(\alpha)}\|_{C^0(K)} \leq C_R \|p - \tilde{p}\|_{C^3(K)}, \quad (48)$$

where C_R depends on $m, N, A, \eta^{-1}, \lambda^{-1}, B_3$. Levi-Civita sectional curvatures also converge on corresponding two-planes whose Gram determinants are uniformly bounded below.

Proof. Differentiating the argument for (38) gives C^1 control of the Christoffel symbols. Insert it into

$$R^\ell_{ijk} = \partial_i \Gamma^\ell_{jk} - \partial_j \Gamma^\ell_{ik} + \Gamma^\ell_{ir} \Gamma^r_{jk} - \Gamma^\ell_{jr} \Gamma^r_{ik}. \quad (49)$$

Sectional curvature is a quotient. The same estimates control its metric numerator, while the uniform Gram bound controls its denominator. $\square$

6.1 Boundary-robust Levi–Civita stability

The preceding estimates are uniform over a probability-coordinate regularity class and are intentionally general in α . For Levi–Civita geometry, the square-root representation gives a sharper result. Define

$$\psi_p(x) = 2(\sqrt{p_1(x)}, \dots, \sqrt{p_N(x)}) \in \mathbb{R}^N. \quad (50)$$

Its image lies on the sphere of radius two. Equip target-valued derivatives with their Euclidean operator norms, and put

$$\|D^j \psi\|_{C^0(K), \text{op}} = \sup_{\substack{x \in K \\ |u_1|, \dots, |u_j| \leq 1}} \|D^j \psi_x[u_1, \dots, u_j]\|_{\ell_2}, \quad (51)$$

and put

$$\delta_j = \sup_{x \in K} \|D^j \psi_p(x) - D^j \psi_{\tilde{p}}(x)\|_{\text{op}}, \quad j = 1, 2. \quad (52)$$

Theorem 6.5 (Vocabulary-independent Levi–Civita stability). *Suppose $\psi_p, \psi_{\tilde{p}} : U \rightarrow \mathbb{R}^N$ are C^2 immersions and, on K ,*

$$\begin{aligned} \|D\psi_p\|_{\text{op}}, \|D\psi_{\tilde{p}}\|_{\text{op}} &\leq B_1, \\ \|D^2\psi_p\|_{\text{op}}, \|D^2\psi_{\tilde{p}}\|_{\text{op}} &\leq B_2, \\ g_p, g_{\tilde{p}} &\succeq \lambda I. \end{aligned} \quad (53)$$

Then

$$\|g_p - g_{\tilde{p}}\|_{C^0(K)} \leq 2B_1\delta_1, \quad (54)$$

$$\|Dg_p - Dg_{\tilde{p}}\|_{C^0(K)} \leq 2(B_1\delta_2 + B_2\delta_1), \quad (55)$$

$$\|\Gamma_p^{\text{LC}} - \Gamma_{\tilde{p}}^{\text{LC}}\|_{C^0(K)} \leq D_{\text{LC}}, \quad (56)$$

where

$$D_{\text{LC}} = \lambda^{-1}(B_1\delta_2 + B_2\delta_1) + 2B_1^2 B_2 \lambda^{-2} \delta_1. \quad (57)$$

For a path γ of Euclidean length L_γ ,

$$\|P_\gamma^{p, \text{LC}} - P_\gamma^{\tilde{p}, \text{LC}}\|_{\text{op}} \leq L_\gamma D_{\text{LC}} \min \left\{ \exp \left(\frac{B_1 B_2}{\lambda} L_\gamma \right), \frac{B_1^2}{\lambda} \right\}. \quad (58)$$

Every displayed constant is independent of N and of $\min_a p_a, \min_a \tilde{p}_a$. The second transport branch uses metric compatibility and prevents an unavoidable exponential-in-length interpretation. It does not remove rank conditioning: after substituting D_{LC} , that branch contains terms proportional to λ^{-2} and λ^{-3} .

Proof. Differentiating (50) gives the exact identities

$$(g_p)_{ij} = \langle \partial_i \psi_p, \partial_j \psi_p \rangle, \quad (\Gamma_p^{\text{LC}})_{ij,k} = \langle \partial_{ij} \psi_p, \partial_k \psi_p \rangle. \quad (59)$$

The first is (7); the second follows either from the Koszul formula or by differentiating the first and cancelling the symmetric terms. Subtracting the two Gram matrices proves (54). Differentiating them once and applying the product rule proves (55). The lower-index coefficients in (59) differ by at most $B_1 \delta_2 + B_2 \delta_1$. Moreover,

$$\|g_p^{-1} - g_{\tilde{p}}^{-1}\|_{\text{op}} \leq 2B_1 \lambda^{-2} \delta_1, \quad (60)$$

and each lower-index coefficient has norm at most $B_1 B_2$. Raising the final index yields (56)–(57). The same identities give $\|\Gamma^{\text{LC}}\|_{\text{op}} \leq B_1 B_2 / \lambda$, so Duhamel’s formula and Gronwall’s inequality give the exponential branch of (58).

For the second branch, note that

$$\lambda I \preceq g_p, g_{\tilde{p}} \preceq B_1^2 I. \quad (61)$$

Levi–Civita transport is an isometry for its own metric; hence every subpath propagator, measured in the background Euclidean norm, has operator norm at most $B_1 / \sqrt{\lambda}$. The exact variation-of-connection identity contains one propagator from each geometry and one copy of their connection defect. Taking norms and integrating $|\dot{\gamma}|$ gives

$$\|P_{\gamma}^{p,\text{LC}} - P_{\gamma}^{\tilde{p},\text{LC}}\|_{\text{op}} \leq L_{\gamma} \frac{B_1^2}{\lambda} D_{\text{LC}}. \quad (62)$$

Taking the better of the two valid estimates proves (58). Appendix A gives the full Duhamel identity and its finite-scale interpretation. $\square$

Corollary 6.6 (Boundary-robust Levi–Civita curvature). *Under the hypotheses of Theorem 6.5, suppose additionally that the two square-root immersions are C^3 so that their classical curvature tensors are defined. Then the covariant Levi–Civita curvature tensors satisfy*

$$\|\text{Rm}_p - \text{Rm}_{\tilde{p}}\|_{C^0(K)} \leq C(B_1, B_2, \lambda^{-1})(\delta_1 + \delta_2). \quad (63)$$

Consequently, sectional curvatures converge on corresponding planes whose Fisher Gram determinants are uniformly bounded below. The constant is again independent of vocabulary size and minimum token probability.

Proof. Regard each square-root map as a Euclidean immersion. Its second fundamental form is

$$\Pi_p(u, v) = (I - \Pi_p) D^2 \psi_p(u, v), \quad \Pi_p = D\psi_p g_p^{-1} (D\psi_p)^*. \quad (64)$$

The inverse-metric estimate used above controls $\Pi_p - \Pi_{\tilde{p}}$ by $C(B_1, \lambda^{-1}) \delta_1$, and hence controls the second fundamental forms by $C(B_1, B_2, \lambda^{-1})(\delta_1 + \delta_2)$. The Euclidean Gauss equation expresses Rm as a quadratic contraction of Π , proving the tensor bound. The sectional-curvature statement follows by controlling its metric denominator. $\square$

6.2 Score moments and the affine-head specialization

The absence of explicit boundary factors above is not an algebraic accident. Let $Y \sim p(x)$ and define the score and its derivative by

$$S_i(Y, x) = \partial_i \log p_Y(x), \quad H_{ij}(Y, x) = \partial_i S_j(Y, x). \quad (65)$$

Proposition 6.7 (Score-moment form of the α -connection). *For every smooth categorical family,*

$$(g_p)_{ij} = \mathbb{E}_p[S_i S_j], \quad (66)$$

$$(C_p)_{ijk} = \mathbb{E}_p[S_i S_j S_k], \quad (67)$$

$$\Gamma_{ij,k}^{(\alpha)} = \mathbb{E}_p[H_{ij} S_k] + \frac{1-\alpha}{2} \mathbb{E}_p[S_i S_j S_k]. \quad (68)$$

For an affine softmax head in a common natural-coordinate chart, let G, C and $\tilde{G}, \tilde{C}$ denote the corresponding second and third centered moments of the decoder rows. Then

$$\Gamma^{(\alpha)} = \frac{1-\alpha}{2} G^{-1} C. \quad (69)$$

If $G, \tilde{G} \succeq \lambda I$, then

$$\begin{aligned} \|\Gamma^{(\alpha)} - \tilde{\Gamma}^{(\alpha)}\|_{\text{op}} &\leq \frac{|1-\alpha|}{2} \left(\lambda^{-1} \|C - \tilde{C}\|_{\text{op}} \right. \\ &\quad \left. + \lambda^{-2} \|G - \tilde{G}\|_{\text{op}} \|\tilde{C}\|_{\text{op}} \right). \end{aligned} \quad (70)$$

This bound has no explicit dependence on vocabulary size or minimum token probability.

Proof. Since $\partial_i p_a = p_a S_i(a)$ and

$$\partial_{ij} p_a = p_a (H_{ij}(a) + S_i(a) S_j(a)), \quad (71)$$

substitution into (7), (8), and (41) (Lemma B.1) proves (66)–(68). At an affine head, $S(Y) = w_Y - \mathbb{E}_p w_Y$ and $H_{ij}(Y) = -G_{ij}$ is independent of Y . Because $\mathbb{E}_p S = 0$, the first term of (68) vanishes, giving (69). Finally use

$$G^{-1} C - \tilde{G}^{-1} \tilde{C} = G^{-1} (C - \tilde{C}) + (G^{-1} - \tilde{G}^{-1}) \tilde{C} \quad (72)$$

and $\|G^{-1} - \tilde{G}^{-1}\|_{\text{op}} \leq \lambda^{-2} \|G - \tilde{G}\|_{\text{op}}$. $\square$

Corollary 6.8 (General α stability from cubic-moment control). *In the setting of Theorem 6.5, suppose additionally that*

$$\|C_p\|_{C^0(K), \text{op}}, \|C_p^\sim\|_{C^0(K), \text{op}} \leq M_3, \quad \|C_p - C_p^\sim\|_{C^0(K), \text{op}} \leq \delta_C. \quad (73)$$

Then

$$\|\Gamma_p^{(\alpha)} - \Gamma_p^{\sim(\alpha)}\|_{\text{op}} \leq D_{\text{LC}} + \frac{|\alpha|}{2} \left(\lambda^{-1} \delta_C + 2B_1 M_3 \lambda^{-2} \delta_1 \right). \quad (74)$$

Thus square-root C^2 control alone determines Levi-Civita stability; a general nonzero- α connection additionally requires stability of the raised third score moment. A tokenwise probability floor is one sufficient route, not a necessary one. Affine softmax structure supplies a floor-free route directly through Proposition 6.7.

Proof. Use $\Gamma^{(\alpha)} = \Gamma^{\text{LC}} - (\alpha/2)g^{-1}C$, Theorem 6.5, and the inverse-metric estimate in its proof. $\square$

Proposition 6.9 (Square-root control does not control nonzero α). *There is a one-dimensional categorical family whose square-root map has uniformly bounded derivatives of every order and whose Fisher metric is identically one, while every fixed nonzero- α connection coefficient diverges toward the simplex boundary.*

Proof. For $0 < \theta < \pi$, take the Bernoulli family

$$p_\theta = \left(\sin^2 \frac{\theta}{2}, \cos^2 \frac{\theta}{2} \right). \quad (75)$$

Its square-root representation is

$$\psi_\theta = 2 \left(\sin \frac{\theta}{2}, \cos \frac{\theta}{2} \right). \quad (76)$$

All derivatives are bounded as $\theta \downarrow 0$, while

$$g_{\theta\theta} = 1, \quad \Gamma_{\theta\theta,\theta}^{\text{LC}} = \langle \psi_\theta'', \psi_\theta' \rangle = 0. \quad (77)$$

The two score values are $\cot(\theta/2)$ and $-\tan(\theta/2)$, so

$$C_{\theta\theta\theta} = 2 \cot \theta. \quad (78)$$

Consequently,

$$\Gamma_{\theta\theta,\theta}^{(\alpha)} = -\alpha \cot \theta, \quad (79)$$

which diverges for every $\alpha \neq 0$. Thus square-root C^2 control and a perfect Fisher spectral floor are sufficient for Levi–Civita stability but not for a general α -connection. At an affine softmax head in natural coordinates, Proposition 6.7 supplies a separate moment-based route and $\Gamma^{(1)} = 0$ identically. $\square$

Corollary 6.10 (Sharp connection selectivity of square-root control). *Within the Amari α -family, $\alpha = 0$ is the unique fixed parameter for which uniform square-root C^2 bounds, square-root C^2 agreement, and a Fisher spectral floor alone imply uniform connection stability in a declared background chart. More precisely, for every fixed $\alpha \neq 0$ there are pairs of Bernoulli predictive maps on the same compact chart such that their square-root maps converge to one another in C^2 , both Fisher metrics are identically one, and yet*

$$\|\Gamma_{p_n}^{(\alpha)} - \Gamma_{\tilde{p}_n}^{(\alpha)}\|_{C^0} \longrightarrow \infty. \quad (80)$$

This is a uniqueness statement only for stability from these stated data; it is not a claim that Levi–Civita is the unique geometrically natural connection.

Proof. The $\alpha = 0$ implication is Theorem 6.5. For the converse, let $K = [0, 1]$, $\varepsilon_n = 1/n$, and use the Bernoulli construction in Proposition 6.9 with angles

$$\theta_n(x) = x + \varepsilon_n, \quad \tilde{\theta}_n(x) = x + 2\varepsilon_n. \quad (81)$$

The associated square-root maps are phase shifts of the same radius-two circle and therefore converge to one another in $C^2(K)$ at rate $O(\varepsilon_n)$; their first and second derivative envelopes are respectively one and one half, and both induced metrics are dx^2 . At $x = 0$ their connection-coefficient defect is

$$|\alpha| |\cot \varepsilon_n - \cot(2\varepsilon_n)| \sim \frac{|\alpha|}{2\varepsilon_n}, \quad (82)$$

which proves (80). $\square$

Remark 6.11 (Why the assumptions matter). The probability-coordinate constants in Theorems 6.2 and 6.4 diverge near $p_a = 0$ because their uniform class controls derivatives of p_a independently of p_a . This is not always an intrinsic blow-up of the pulled-back geometry: Theorem 6.5 captures cancellations through square-root derivatives, while Proposition 6.7 captures them through probability-weighted score moments. Rank loss remains intrinsic because raising indices requires g^{-1} . For nonzero α , the raised third score moment supplies information not determined by the Fisher metric alone and must be controlled explicitly; uniform interiority is sufficient but not logically necessary. Vocabulary independence means that no explicit N occurs in the displayed constants; the derivative bounds and spectral floor must still be uniform and may deteriorate with model scale. The estimates are conditional finite-scale bounds, not numerical certificates until those quantities are measured.

7 From cross-entropy to transport under Sobolev regularity

Theorem 6.2 assumes derivative-level agreement. We now give sufficient conditions under which population KL convergence implies it.

Let Ω be a compact m -dimensional smooth manifold or a bounded Sobolev-extension domain. Fix a smooth background Riemannian metric on the former and the Euclidean metric on the latter; all volume measures, Sobolev norms, operator norms, and path lengths in this section refer to that background structure. Let μ have density ρ with respect to volume, satisfying $\rho \geq \rho_0 > 0$. Define

$$\mathcal{K}(p, \tilde{p}) = \int_{\Omega} D_{\text{KL}}(p(x) \| \tilde{p}(x)) \, d\mu(x). \quad (83)$$

Theorem 7.1 (Risk-to-transport convergence). *Let $p, \tilde{p} : \Omega \rightarrow \Delta_N^{\circ}$ satisfy the probability and spectral floors of Assumption 6.1. Suppose*

$$p, \tilde{p} \in H^s(\Omega; \mathbb{R}^N), \quad s > 2 + \frac{m}{2}, \quad (84)$$

and

$$\|p\|_{H^s} + \|\tilde{p}\|_{H^s} \leq B. \quad (85)$$

Fix $A, L_0 > 0$. For every r with

$$2 + \frac{m}{2} < r < s, \quad (86)$$

there is a constant $C = C(\Omega, m, s, r, N, A, L_0, \rho_0^{-1}, \eta^{-1}, \lambda^{-1}, B)$ such that, uniformly for $|\alpha| \leq A$ and every path γ of background length $L_{\gamma} \leq L_0$,

$$\|P_{\gamma}^{p, \alpha} - \tilde{P}_{\gamma}^{\tilde{p}, \alpha}\|_{\text{op}} \leq C B^{r/s} \mathcal{K}(p, \tilde{p})^{\frac{s-r}{2s}}. \quad (87)$$

Equivalently, the transport error is $O(\mathcal{K}^{\beta})$ for every

$$0 < \beta < \frac{s - 2 - m/2}{2s}. \quad (88)$$

Proof. Pinsker's inequality and $\rho \geq \rho_0$ give

$$\begin{aligned} \|p - \tilde{p}\|_{L^2(\text{dvol})}^2 &\leq \rho_0^{-1} \int_{\Omega} \|p - \tilde{p}\|_1^2 \, d\mu \\ &\leq 2\rho_0^{-1} \mathcal{K}(p, \tilde{p}). \end{aligned} \quad (89)$$

Sobolev interpolation [1] yields

$$\|p - \tilde{p}\|_{H^r} \leq C \|p - \tilde{p}\|_{L^2}^{1-r/s} \|p - \tilde{p}\|_{H^s}^{r/s}, \quad (90)$$

and therefore

$$\|p - \tilde{p}\|_{H^r} \leq C B^{r/s} \mathcal{K}(p, \tilde{p})^{(s-r)/(2s)}. \quad (91)$$

Since $r > 2 + m/2$, Sobolev embedding gives $H^r \hookrightarrow C^2$. Apply Theorem 6.2, which is stated on a Euclidean chart; when Ω is a manifold, use its atlas form, Lemma B.16. $\square$

Theorem 7.2 (Boundary-robust risk-to-Levi-Civita transport). *Let $p, \tilde{p} : \Omega \rightarrow \Delta_N^\circ$, write $\psi_p = 2\sqrt{p}$ and $\psi_{\tilde{p}} = 2\sqrt{\tilde{p}}$ componentwise, and assume*

$$\psi_p, \psi_{\tilde{p}} \in H^s(\Omega; \ell_2^N), \quad s > 2 + \frac{m}{2}, \quad (92)$$

with

$$\|\psi_p\|_{H^s} + \|\psi_{\tilde{p}}\|_{H^s} \leq B, \quad g_p, g_{\tilde{p}} \succeq \lambda I. \quad (93)$$

Fix $L_0 > 0$. For every r satisfying $2 + m/2 < r < s$ and every path of background length at most L_0 ,

$$\|P_\gamma^{p, \text{LC}} - P_\gamma^{\tilde{p}, \text{LC}}\|_{\text{op}} \leq C \mathcal{K}(p, \tilde{p})^{\frac{s-r}{2s}}, \quad (94)$$

where C depends on $\Omega, s, r, \rho_0^{-1}, B, \lambda^{-1}, L_0$ but not on N or a minimum token probability. Equivalently, the transport error is $O(\mathcal{K}^\beta)$ for every

$$0 < \beta < \frac{s - 2 - m/2}{2s}. \quad (95)$$

Proof. For categorical distributions Lemma B.4 gives the scalar inequality

$$\sum_a (\sqrt{p_a} - \sqrt{\tilde{p}_a})^2 \leq D_{\text{KL}}(p \| \tilde{p}) \quad (96)$$

gives

$$\|\psi_p - \psi_{\tilde{p}}\|_{L^2(\text{dvol}; \ell_2)}^2 \leq 4\rho_0^{-1} \mathcal{K}(p, \tilde{p}). \quad (97)$$

Hilbert-valued Sobolev interpolation and embedding, with constants independent of the target dimension (Lemmas B.12–B.14), give

$$\begin{aligned} \|\psi_p - \psi_{\tilde{p}}\|_{C^2(\ell_2)} &\leq C \|\psi_p - \psi_{\tilde{p}}\|_{H^r(\ell_2)} \\ &\leq C B^{r/s} \mathcal{K}(p, \tilde{p})^{(s-r)/(2s)}. \end{aligned} \quad (98)$$

The same embedding bounds the first two derivatives of each square-root map in terms of B . Apply Theorem 6.5. Remark B.15 tracks where each constant comes from and confirms that no step introduces a factor of N or of $\min_a p_a$. $\square$

Corollary 7.3 (Two-model geometric convergence). *Suppose two model sequences $p_{A,n}, p_{B,n}$ on a shared intervention chart converge in population cross-entropy to the same conditional map p_* , and all three maps satisfy uniform versions of the hypotheses of Theorem 7.1. Then, for every admissible β ,*

$$\|P_\gamma^{A,n,\alpha} - P_\gamma^{B,n,\alpha}\|_{\text{op}} \leq C(\epsilon_{A,n}^\beta + \epsilon_{B,n}^\beta), \quad (99)$$

where $\epsilon_{i,n} = \mathbb{E}D_{\text{KL}}(p_* \| p_{i,n})$. If the models instead live on aligned charts, the same statement holds for the transport commutator in (45) after replacing $p_{B,n}$ by $p_{B,n} \circ \Phi_n$, provided these pulled-back maps converge to the same p_* and satisfy the same uniform hypotheses.

Alternatively, if the square-root maps satisfy uniform versions of Theorem 7.2, their Levi–Civita transports obey the same conclusion with constants independent of vocabulary size and minimum token probability.

Proof. Let $P_\gamma^{*,\alpha}$ denote the α -transport induced by p_* . For $i \in \{A, B\}$, Theorem 7.1 and the uniform hypotheses give

$$\|P_\gamma^{i,n,\alpha} - P_\gamma^{*,\alpha}\|_{\text{op}} \leq C_i \epsilon_{i,n}^\beta. \quad (100)$$

The constants are uniformly bounded in n and may be replaced by one constant C . The operator-norm triangle inequality, applied to the common target transport rather than to the KL divergences, now gives

$$\begin{aligned} \|P_\gamma^{A,n,\alpha} - P_\gamma^{B,n,\alpha}\|_{\text{op}} &\leq \|P_\gamma^{A,n,\alpha} - P_\gamma^{*,\alpha}\|_{\text{op}} + \|P_\gamma^{*,\alpha} - P_\gamma^{B,n,\alpha}\|_{\text{op}} \\ &\leq C(\epsilon_{A,n}^\beta + \epsilon_{B,n}^\beta). \end{aligned} \quad (101)$$

For an alignment Φ_n , work on model A’s chart with the pulled-back predictive map $p_{B,n} \circ \Phi_n$. Exact naturality identifies its transport with

$$(\text{d}\Phi_n)_{\gamma(1)}^{-1} P_{\Phi_n \circ \gamma}^{B,n,\alpha} (\text{d}\Phi_n)_{\gamma(0)}. \quad (102)$$

The same triangle argument therefore gives the commutator estimate in the aligned norms of Corollary 6.3; an independently chosen target background norm incurs the additional alignment factor stated there. Replacing Theorem 7.1 by Theorem 7.2 proves the Levi–Civita alternative. $\square$

If $s > 3 + m/2$, the same argument combined with Theorem 6.4 gives curvature convergence for every

$$0 < \beta < \frac{s - 3 - m/2}{2s}. \quad (103)$$

For Levi–Civita curvature, Corollary 6.6 instead uses C^2 convergence of the square-root immersions. Thus Theorem 7.2 also gives vocabulary-independent LC curvature convergence at the transport exponent, provided additionally that the square-root immersions are C^3 so their classical curvature tensors are defined and that the compared plane Gram determinants stay uniformly positive.

The uniform Sobolev hypothesis excludes the high-frequency behavior in Proposition 5.1. For language models it should be tested on low-dimensional continuous intervention charts, such as soft-prompt paths or controlled activation subspaces, rather than assumed on the discrete set of tokenized contexts.

8 The semantic-sufficiency obstruction

Even perfect agreement of predictive geometry does not guarantee that a semantic operation is a field on that geometry. We now separate this representational obstruction from connection mismatch.

Let $F : \mathcal{H} \rightarrow P$ be a regular predictive quotient, let H be a random hidden state with law μ , and let $X_T(H) \in T_H \mathcal{H}$ describe a semantic operation T . Its observable infinitesimal predictive effect is

$$Z_T(H) = \text{d}F_H X_T(H) \in T_{F(H)} P. \quad (104)$$

Assume square integrability in the Fisher metric. Since P is embedded in the simplex, tangent-valued conditional expectations may be taken fiberwise in the ambient zero-sum space; conditioning on $F(H) = p$ fixes both $T_p P$ and g_p .

Theorem 8.1 (Semantic conditional-variance obstruction). *For a measurable, square-integrable field $Y : P \rightarrow TP$, define*

$$\mathcal{E}_T(Y) = \mathbb{E} \|Z_T(H) - Y(F(H))\|_{g_{F(H)}}^2. \quad (105)$$

Let

$$\bar{Z}_T(p) = \mathbb{E}[Z_T(H) \mid F(H) = p]. \quad (106)$$

Then

$$\begin{aligned} \mathcal{E}_T(Y) &= \mathbb{E} \|Z_T - \bar{Z}_T(F(H))\|_g^2 \\ &\quad + \mathbb{E} \|\bar{Z}_T(F(H)) - Y(F(H))\|_g^2. \end{aligned} \quad (107)$$

Consequently,

$$\inf_Y \mathcal{E}_T(Y) = \mathbb{E} \text{Var}_g(Z_T \mid F). \quad (108)$$

The semantic effect descends almost surely to a predictive field if and only if this conditional variance is zero. Smooth descent additionally requires a smooth version of $\bar{Z}_T$.

Proof. Square-integrable tangent-valued random variables form a Hilbert direct integral with inner product

$$\langle U, V \rangle = \mathbb{E}[g_{F(H)}(U, V)]. \quad (109)$$

By Lemma B.17, conditional expectation maps this space into the $\sigma(F)$ -measurable sections and is the orthogonal projection onto them; in particular that subspace is closed. Equivalently, decompose

$$\begin{aligned} Z_T - Y(F) &= (Z_T - \bar{Z}_T(F)) \\ &\quad + (\bar{Z}_T(F) - Y(F)). \end{aligned} \quad (110)$$

The two terms are orthogonal after conditioning on F , which proves (107) and (108). $\square$

Corollary 8.2 (Injective maps have no quotient obstruction). *If F is injective on the support of H and its inverse on $F(\text{supp } H)$ is measurable, then*

$$\text{Var}_g(Z_T \mid F) = 0 \quad \text{almost surely.} \quad (111)$$

For the affine softmax decoder, this applies whenever

$$\mathcal{N} = \{u : Wu \in \text{span}\{\mathbf{1}\}\} = \{0\}, \quad (112)$$

equivalently whenever the pullback Fisher matrix (14) is positive definite.

Proof. Under the stated injectivity, $H = F^{-1}(F(H))$ almost surely, so every measurable function of H , including $Z_T(H)$, is measurable with respect to $\sigma(F(H))$. Its conditional variance is therefore zero. The affine-softmax equivalence follows from

$$F(h) = F(h') \iff W(h - h') \in \text{span}\{\mathbf{1}\}. \quad (113)$$

$\square$

Corollary 8.2 is a scope condition, not a defect of Theorem 8.1. The obstruction is informative at a pre-quotient representation with nontrivial predictive fibers or at an earlier layer whose remaining network is noninjective. It is identically inert at a minimal full-rank affine head. If a smooth coarsening $S : P \rightarrow \bar{P}$ is declared, the theorem applies anew to $\bar{F} = S \circ F$, the effect $d\bar{F} X_T$, and the metric and tangent bundle of the coarsened image. Merely conditioning the original P -tangent field on $S(F)$ can mix distinct tangent spaces and is not covered by the theorem. Approximate conditioning or finite bins likewise define a different, resolution-dependent diagnostic.

Corollary 8.3 (Connection-restricted decomposition). *Let $\mathcal{C}_\alpha$ be any closed class of predictive fields selected by an α -connection, for example fields satisfying $\nabla^{(\alpha)} Y = 0$ on a specified simply connected chart or fields with a bounded covariant-derivative residual. Then*

$$\inf_{Y \in \mathcal{C}_\alpha} \mathcal{E}_T(Y) = \underbrace{\mathbb{E} \text{Var}_g(Z_T | F)}_{\text{predictive insufficiency}} + \underbrace{\inf_{Y \in \mathcal{C}_\alpha} \mathbb{E} \|\bar{Z}_T - Y\|_g^2}_{\text{connection mismatch}}. \quad (114)$$

Proof. Equation (107) holds for every $Y \in \mathcal{C}_\alpha$, and its conditional-variance term is independent of Y . Taking the infimum over $\mathcal{C}_\alpha$ gives (114). $\square$

Corollary 8.4 (Cross-model semantic transfer). *Let P_A, P_B be predictive quotients related by a C^1 diffeomorphism $\Phi : P_A \rightarrow P_B$ on the relevant supports. Write $Q_i = F_i(H_i)$ for the random predictive state of model i . Define*

$$Y_{A \rightarrow B}(p_B) = d\Phi_{\Phi^{-1}(p_B)} \bar{Z}_A(\Phi^{-1}(p_B)). \quad (115)$$

Then

$$\begin{aligned} \mathbb{E} \|Z_B - Y_{A \rightarrow B}(Q_B)\|_{g_B}^2 &= \underbrace{\mathbb{E} \|Z_B - \bar{Z}_B(Q_B)\|_{g_B}^2}_{\text{model-B quotient obstruction}} \\ &+ \underbrace{\mathbb{E} \|\bar{Z}_B - d\Phi \bar{Z}_A\|_{g_B}^2}_{\text{cross-model field mismatch}}. \end{aligned} \quad (116)$$

Proof. Apply Theorem 8.1 to model B with the measurable predictive field $Y = Y_{A \rightarrow B}$. Its first term is model B's conditional Fisher variance, and its second term is

$$\mathbb{E} \|\bar{Z}_B(Q_B) - Y_{A \rightarrow B}(Q_B)\|_{g_B}^2 = \mathbb{E} \|\bar{Z}_B - d\Phi \bar{Z}_A\|_{g_B}^2, \quad (117)$$

where the final expression evaluates $\bar{Z}_A$ at $\Phi^{-1}(Q_B)$. This is (116). $\square$

Identity (116) is measured in g_B and requires only that the transferred field be well defined and measurable. If Φ is additionally a Fisher isometry, the mismatch has an intrinsic metric-preserving interpretation. Otherwise it is relative to the proposed alignment and its distortion must be reported separately.

No choice of α can reduce the first term of (114). On a genuinely noninjective predictive map, a strictly positive population conditional Fisher variance therefore proves that no predictive-state-only field, and hence no choice of α -transport on that quotient, can represent the specified infinitesimal effect exactly. At an injective final head the term vanishes tautologically and is not an empirical diagnostic.

9 Operational tests and training implications

The theorems suggest a hierarchy of increasingly strong cross-model agreement tests:

$$\begin{aligned}
\text{Level 0} & : \text{ output KL agreement,} \\
\text{Level 1} & : \text{ Fisher metric agreement,} \\
\text{Level 2} & : \alpha\text{-connection and transport agreement,} \\
\text{Level 3} & : \text{ semantic-field equivariance,} \\
\text{Level 4} & : \text{ curvature and holonomy agreement.}
\end{aligned} \tag{118}$$

Agreement at one level does not imply the next without the regularity and sufficiency conditions identified above.

9.1 Transport commutation score

For an estimated alignment Φ and sampled paths and tangent vectors, define

$$\mathcal{D}_{\text{PT}}^{(\alpha)}(A, B; \Phi) = \mathbb{E}_{\gamma, v} \frac{\|d\Phi_{\gamma(1)} P_{\gamma}^{A, \alpha} v - P_{\Phi\gamma}^{B, \alpha} d\Phi_{\gamma(0)} v\|_{g_B(\Phi(\gamma(1)))}}{\|d\Phi_{\gamma(0)} v\|_{g_B(\Phi(\gamma(0)))}}. \tag{119}$$

The numerator is measured in model B 's Fisher norm at the endpoint and normalized by the aligned input in model B 's Fisher norm at the base point. Since Φ is a local diffeomorphism, the denominator is nonzero for $v \neq 0$. This relative score is invariant under separate reparameterizations of either chart and does not decrease merely because every alignment derivative is multiplied by a common scalar. Because Φ also changes the compared path, alignment scale and distortion must still be audited. Define

$$L(\Phi) = \sup_{x \in K} \|d\Phi_x\|_{g_A \rightarrow g_B}, \quad L(\Phi^{-1}) = \sup_{x \in K} \|d\Phi_x^{-1}\|_{g_B \rightarrow g_A}, \quad \kappa(\Phi) = L(\Phi)L(\Phi^{-1}), \tag{120}$$

and report all three quantities rather than κ alone: uniform rescaling changes the two Lipschitz factors inversely while leaving their product unchanged. Alignment fitting must impose a preregistered scale or bi-Lipschitz range, and scores computed under different unconstrained alignments are not comparable. Theorem 4.1 makes this score zero under exact predictive equivalence. Corollary 6.3 bounds its aligned-norm counterpart under approximate equivalence; converting that bound to the relative Fisher score incurs the declared norm-equivalence and lower-singular-value factors. A practical study should test whether (119) predicts held-out steering transfer or task generalization beyond output KL and representation-only similarities such as CKA.

9.2 Selecting the connection

At a linear softmax head, $\alpha = 1$ is ordinary vector reuse, $\alpha = 0$ is Fisher-metric-compatible Levi-Civita transport, and $\alpha = -1$ is mixture transport. A predictive semantic field Y_T can be evaluated through

$$\mathcal{R}_T(\alpha) = \mathbb{E} \|\nabla^{(\alpha)} Y_T\|_g^2. \tag{121}$$

The minimizer is interpreted relative to the tested field and chart:

- a minimum at the exponential connection means that the tested field is closest to exponential-parallel; at the affine softmax head this is constant natural-coordinate vector reuse;
- a minimum at the Levi-Civita connection means that the tested field is closest to the metric-compatible transport;

- a minimum at the mixture or an intermediate α identifies the corresponding affine connection as the closest member of the tested family; intermediate α -connections need not be flat;
- on a verified noninjective predictive map, positive population conditional variance proves predictive insufficiency independently of the minimizing connection;
- on such a map, small estimated conditional variance together with large residuals for every tested α motivates an operation-specific connection or a nonparallel field model.

The construction of the tested field must be declared separately from the connection comparison. A finite hidden difference $h(Tc) - h(c)$ is the exponential logarithm of its endpoints at an affine head and therefore intentionally tests transfer of the conventional exponential-coordinate intervention. A connection-neutral infinitesimal test instead specifies a continuous intervention T_t and uses

$$X_T(h(c)) = \left. \frac{d}{dt} \right|_{t=0} h(T_t c). \quad (122)$$

For finite endpoints, each candidate can be evaluated with its own $\text{Log}^{(\alpha)}\text{-transport-Exp}^{(\alpha)}$ construction. Any minimizing $\alpha \neq 1$ remains a descriptive best fit within the tested family; it does not by itself identify a mechanism implemented by optimization.

9.3 Training regimes

The risk-to-transport theorems indicate why derivative regularity matters. A causal training study can compare cross-entropy-only models with matched-loss models trained using

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \widehat{\text{Var}}_g(dFX_T | F) + \mu \|\nabla^{(\alpha)} \bar{Z}_T\|_g^2 + \nu \mathcal{R}_{\text{smooth}}. \quad (123)$$

The first regularizer is included only for a declared noninjective predictive map and encourages the operation to descend through that map. Under coarsening, every occurrence of F , dFX_T , and g in this term must be replaced by $\bar{F} = S \circ F$, $d\bar{F}X_T$, and the coarsened-image metric. It must be omitted at an injective final head, where it is identically zero. The second selects connection-relative consistency. For Levi-Civita transport, the final term should control the square-root predictive map in the Sobolev norm used below; for general α , it must additionally control the required score moments. Applying either theorem requires verifying the corresponding uniform bound. The empirical protocol treats behavioral improvement as the primary outcome and the geometric regularizers as diagnostic quantities.

A no-GPU pilot can use frozen small models and one- or two-dimensional soft-prompt charts. Finite differences estimate F , ∂F , and $\partial^2 F$; the model's output head provides exact vocabulary probabilities. The main comparisons are output KL, metric defect, the transport score (119), connection residual, and held-out intervention transfer. Conditional variance is added only at a representation with verified nontrivial predictive fibers or after a complete coarsened map, effect, and image geometry are explicitly defined.

An inferential curvature study must also separate semantic structure from Fisher-spectrum alignment. Isotropic Fisher-Haar planes answer a literal isotropic null but are not spectrum matched. A task-specific analysis should preserve the semantic plane's leverage across preregistered Fisher spectral bands, using signs for singleton bands and basis-invariant Haar rotations inside repeated or clustered bands. Fine relative-gap bands can reduce the control orbit mostly to sign flips, so the analysis must be repeated with preregistered coarser relative-log bands and conclusions that change with the band rule must be labeled control-sensitive. A plus-one control-tail rank is a calibrated

randomization p -value only under the explicit conditional invariance null for its declared block-orthogonal group; direction and multiplicity must also be preregistered. First-order α fits must be extrapolated to zero edge length or replaced by integrated transport; their finite-edge parameter error is generically first order in Fisher length.

10 Limitations

Continuous charts. Tokenized contexts are discrete. Derivatives and connections require an explicit continuous chart, such as soft prompts, activation interventions, or a justified local manifold model.

Boundary and rank. Softmax probabilities are positive but can be extremely small. Uniform tokenwise floors are mathematically clean and practically strong, and the general- α probability-coordinate bound is not a useful finite-scale certificate when its constants are dominated by that floor. The square-root Levi-Civita theorem removes this explicit dependence, and affine heads admit score-moment formulas, but Fisher rank loss remains a genuine instability. In particular, the metric-compatible transport branch contains λ^{-2} and λ^{-3} terms after D_{LC} is expanded. The historical pilot’s pointwise spectral floors worsen by more than two orders of magnitude between initialization and the final checkpoint, causing much larger inverse-power factors even before the unmeasured derivative envelopes are inserted; Appendix A records the values. Thus vocabulary independence is not a claim of a nonvacuous trained-model certificate. Quotienting, coarse-graining, or a preregistered horizontal-subspace analysis is necessary near rank loss; an unreported ridge changes the geometry.

Architectural support. The ambient affine-head family permits arbitrary interventions in its chosen hidden coordinates. Natural post-LayerNorm states satisfy architectural constraints and need not fill that family or remain on their support after vector addition. Claims about natural-state geometry therefore require a chart tangent to the reachable support or a pullback through the normalization map; ambient intervention geometry and support-intrinsic geometry must not be conflated.

Different tokenizers. The exact theorem assumes a common outcome space. A tokenizer map generally acts as a Markov morphism. Fisher geometry is preserved only under a statistically sufficient or reversible identification; otherwise it contracts [5].

Population versus empirical loss. Equation (25) concerns population risk, realizability, and global optimization. Finite empirical loss introduces estimation and optimization errors not treated here.

Alignment is not identified automatically. The stability theorem evaluates a proposed Φ . Constructing a statistically identifiable alignment from finite observations, especially across different latent dimensions, is a separate problem. One may instead work on a shared low-dimensional intervention chart.

Connection dependence. Parallel transport compares tangent vectors at different points relative to a selected connection. At the final affine softmax head, ambient addition coincides with

exponential-affine transport, while Levi-Civita and mixture transport preserve different structures. The theory therefore evaluates connection choice separately across operations, layers, and tasks.

11 Conclusion

Idealized pretraining makes convergence of conditional output distributions natural because excess cross-entropy is expected KL divergence. It does not by itself make convergence of Fisher metrics, affine connections, or curvature natural. Geometry depends on derivatives, and a regular oscillatory counterexample separates these notions.

Under explicit smoothness and identifiability assumptions, stronger conclusions become provable. Predictive maps that converge in C^2 have convergent Amari α -connections and parallel transports; C^3 convergence controls curvature. For Levi-Civita transport, square-root coordinates give vocabulary-independent bounds without a uniform probability floor, and integrated KL plus a uniform Hilbert-valued Sobolev bound supplies the missing derivative convergence. Nonzero- α connections require additional raised-third-score-moment control, for which a tokenwise probability floor is sufficient but not necessary. The semantic conditional-variance obstruction determines whether a noninjective predictive map is rich enough to support the operation; at an injective head it vanishes by construction.

If aligned predictive maps have vanishing population KL risk together with the uniform Sobolev, boundary, and rank bounds of Theorem 7.1, their α -transport commutators vanish. Semantic-field equivariance is a separate condition: it requires small within-model quotient variance and small cross-model conditional-mean-field mismatch. These quantities localize deviations from transfer to inadequate regularity, predictive insufficiency, field mismatch, or connection mismatch.

A Square-root boundary calculus and finite-scale auditing

A.1 What the square-root sphere does and does not flatten

The classical square-root map $\psi = 2\sqrt{p}$ sends the open categorical simplex to the positive orthant of the radius-two sphere and satisfies

$$\sum_a \psi_a^2 = 4, \quad g = D\psi^* D\psi. \quad (124)$$

The ambient inner product is Euclidean, but the induced Fisher geometry is not globally Euclidean: the full simplex has Levi-Civita sectional curvature $1/4$. Thus the sphere representation supplies a Gram factorization of the metric, not a proof of intrinsic flatness.

The metric and first-kind Levi-Civita coefficients,

$$g_{ij} = \langle \partial_i \psi, \partial_j \psi \rangle, \quad \Gamma_{ij,k}^{\text{LC}} = \langle \partial_{ij} \psi, \partial_k \psi \rangle, \quad (125)$$

are polynomial contractions of first and second immersion derivatives. Raised coefficients and tangent projections still contain g^{-1} , so the square-root representation removes explicit tokenwise denominators but does not remove Fisher rank conditioning.

For the cubic tensor, $p_a = \psi_a^2/4$ and $\partial_i p_a = (\psi_a/2)\partial_i \psi_a$ give the exact identity

$$C_{ijk} = 2 \sum_a \frac{\partial_i \psi_a \partial_j \psi_a \partial_k \psi_a}{\psi_a}. \quad (126)$$

Consequently, square-root C^2 bounds alone do not uniformly bound C near a vanishing component. Equation (126) does not say that C must diverge: its numerator can vanish sufficiently quickly, or structural score-moment identities can provide cancellation.

Proposition A.1 (Nonzero- α stability criterion). *On a common chart, put $K_p = g_p^{-1}C_p$ and $K_{\tilde{p}} = g_{\tilde{p}}^{-1}C_{\tilde{p}}$. For every fixed $\alpha \neq 0$,*

$$\|\Gamma_p^{(\alpha)} - \Gamma_{\tilde{p}}^{(\alpha)}\| \leq \|\Gamma_p^{\text{LC}} - \Gamma_{\tilde{p}}^{\text{LC}}\| + \frac{|\alpha|}{2} \|K_p - K_{\tilde{p}}\|, \quad (127)$$

$$\|K_p - K_{\tilde{p}}\| \leq \frac{2}{|\alpha|} \left(\|\Gamma_p^{(\alpha)} - \Gamma_{\tilde{p}}^{(\alpha)}\| + \|\Gamma_p^{\text{LC}} - \Gamma_{\tilde{p}}^{\text{LC}}\| \right). \quad (128)$$

Hence, once Levi-Civita stability is known, stability of one fixed nonzero- α connection is equivalent to stability of the raised third score moment. No tokenwise probability floor is logically necessary.

Proof. Both estimates are the triangle inequality applied in opposite directions to

$$\Gamma^{(\alpha)} = \Gamma^{\text{LC}} - \frac{\alpha}{2} K. \quad (129)$$

□

At an affine softmax head, write

$$S_a(u) = (w_a - \mathbb{E}_p w_Y)^\top u. \quad (130)$$

Then the first two derivatives of the square-root map satisfy

$$D\psi_a[u] = \sqrt{p_a} S_a(u), \quad (131)$$

$$D^2\psi_a[u, v] = \sqrt{p_a} \left(\frac{1}{2} S_a(u) S_a(v) - g(u, v) \right). \quad (132)$$

Because $\mathbb{E}_p[S(u)S(v)] = g(u, v)$, expansion of the squared norm gives

$$\|D^2\psi[u, v]\|_{\ell_2}^2 = \frac{1}{4} \mathbb{E}_p[S_Y(u)^2 S_Y(v)^2]. \quad (133)$$

By Cauchy-Schwarz, the associated bilinear operator norm has the exact pointwise form

$$\|D^2\psi_x\|_{\text{op}} = \frac{1}{2} \left(\sup_{|u| \leq 1} \mathbb{E}_{p(x)}[S_Y(u)^4] \right)^{1/2}. \quad (134)$$

Indeed, the mixed fourth moment is at most the geometric mean of the two pure fourth moments, while choosing $u = v$ at a maximizer attains the supremum. Thus the missing uniform B_2 envelope is exactly one half of the square root of the worst fourth score moment over chart-unit directions and points on the declared domain. Finite directional probes estimate this quantity but do not certify the supremum. Proposition 6.7 separately supplies the third moment needed by nonzero- α connections and the exact cancellation $\Gamma^{(1)} = 0$ in natural affine-head coordinates.

The Bernoulli family in Proposition 6.9 proves that the extra condition cannot be discarded in general. More invariantly, because $g = d\theta^2$ there,

$$\|\nabla^{(\alpha)} - \nabla^{\text{LC}}\|_g = |\alpha \cot \theta| \longrightarrow \infty. \quad (135)$$

Flat exponential or mixture coordinates do not contradict this statement: their coordinate changes become nonuniform near the boundary. The main theorems concern interior maps with no *uniform* probability floor. If exact zeros or changing support are admitted, smoothness of ψ must be imposed directly and the induced object is a Hellinger extension rather than an ordinary regular Fisher model on the open simplex.

A.2 Metric-compatible transport bound

Let $A = \nabla^{g, \text{LC}} - \nabla^{\tilde{g}, \text{LC}}$. Up to the sign convention for A , the exact variation-of-connection identity is

$$P_{1 \leftarrow 0}^{g, \text{LC}} - P_{1 \leftarrow 0}^{\tilde{g}, \text{LC}} = - \int_0^1 P_{1 \leftarrow t}^{g, \text{LC}} A_{\gamma(t)}(\dot{\gamma}(t), P_{t \leftarrow 0}^{\tilde{g}, \text{LC}} \cdot) dt. \quad (136)$$

The conditioning factors in a background Euclidean norm can be separated from the invariant transport statement. For two Riemannian metrics g and $\tilde{g}$, define the asymmetric mixed norm of $A = \nabla^{g, \text{LC}} - \nabla^{\tilde{g}, \text{LC}}$ by

$$\|A_x\|_{\tilde{g}, \tilde{g} \rightarrow g} = \sup_{\substack{\|u\|_{g_x} \leq 1 \\ \|v\|_{g_x} \leq 1}} \|A_x(u, v)\|_{g_x}. \quad (137)$$

Proposition A.2 (Intrinsic mixed-norm transport comparison). *Along every piecewise C^1 path $\gamma : [0, 1] \rightarrow K$,*

$$\|P_{1 \leftarrow 0}^{g, \text{LC}} - P_{1 \leftarrow 0}^{\tilde{g}, \text{LC}}\|_{\tilde{g}_{\gamma(0)} \rightarrow g_{\gamma(1)}} \leq \int_0^1 \|A_{\gamma(t)}\|_{\tilde{g}, \tilde{g} \rightarrow g} \|\dot{\gamma}(t)\|_{\tilde{g}_{\gamma(t)}} dt. \quad (138)$$

The reverse asymmetric estimate follows by interchanging g and $\tilde{g}$.

Proof. Apply the Duhamel identity (136) to an input z with $\|z\|_{\tilde{g}_{\gamma(0)}} \leq 1$. The inner $\tilde{g}$ -transport preserves its $\tilde{g}$ -norm, and the outer g -transport preserves the g -norm of the output of A . The definition of the mixed norm bounds the remaining contraction pointwise; integration and the supremum over z give (138). $\square$

Equation (138) is coordinate invariant and contains no separate condition-number multiplier. It does not remove conditioning: instability of raising indices or approaching rank loss is absorbed into the mixed norm of A . The Euclidean certificate below is therefore useful when a shared background chart is operationally meaningful, while the mixed-norm form is the cleaner geometric comparison.

Metric compatibility and $\lambda I \preceq g \preceq B_1^2 I$ imply

$$\|P_{t \leftarrow s}^{\text{LC}}\|_{\text{op}} \leq \frac{B_1}{\sqrt{\lambda}} \quad (139)$$

for either geometry. Substitution into (136) gives the polynomial branch of (58).

For finite-scale reporting define

$$\tau_\gamma = \frac{B_1 B_2}{\lambda} L_\gamma, \quad \kappa_{\text{env}} = \frac{B_1^2}{\lambda}. \quad (140)$$

The transport theorem reads

$$\|P_\gamma^{p, \text{LC}} - P_\gamma^{\tilde{p}, \text{LC}}\|_{\text{op}} \leq L_\gamma D_{\text{LC}} \min\{e^{\tau_\gamma}, \kappa_{\text{env}}\}. \quad (141)$$

At one point, $\|D\psi\|_{\text{op}}^2 = \lambda_{\max}(g)$, so pointwise-tight constants give $B_1^2/\lambda = \kappa(g)$. Uniform theorem constants instead give

$$\kappa_{\text{env}} = \frac{\sup_K \lambda_{\max}(g)}{\inf_K \lambda_{\min}(g)} \geq \sup_K \kappa(g), \quad (142)$$

and equality need not hold. Both quantities are relative to the declared background chart. Expanding the metric-compatible branch makes its rank dependence explicit:

$$L_\gamma \frac{B_1^2}{\lambda} D_{\text{LC}} = L_\gamma \left[\frac{B_1^2(B_1\delta_2 + B_2\delta_1)}{\lambda^2} + \frac{2B_1^4 B_2\delta_1}{\lambda^3} \right]. \quad (143)$$

Consequently, removing explicit vocabulary and minimum-probability factors does not by itself make the Euclidean certificate informative near rank loss.

The historical pilot artifact `results/pythia_14m_pilot.json` gives the following pointwise base-state diagnostic in one fixed declared chart. The inverse powers use the smallest observed eigenvalue at each checkpoint and are not substitutes for the uniform pathwise floor required by the theorem.

Checkpoint	observed $\lambda_{\min}$ range	worst λ^{-1}	worst λ^{-2}	worst λ^{-3}
initialization	$[2.770, 2.778] \times 10^{-3}$	3.61×10^2	1.30×10^5	4.71×10^7
step143000	$[1.199, 1.391] \times 10^{-5}$	8.34×10^4	6.96×10^9	5.80×10^{14}

Relative to initialization, the worst displayed inverse, inverse-square, and inverse-cube factors grow by approximately 2.31×10^2 , 5.34×10^4 , and 1.23×10^7 , respectively. These comparisons are chart dependent but meaningful for this fixed-chart diagnostic. They show why a trained-model certificate can be numerically vacuous even when the theorem is correct.

The pilot does not report B_2 , δ_1 , δ_2 , or pathwise extrema, so no complete numerical value of (141) is justified by the committed artifacts. The spectra motivate a conditioning audit but cannot by themselves imply a threshold such as $B_2 L_\gamma \lesssim 10^{-4}$ or establish that the theorem is intrinsically short-path. A numerical certificate requires measuring every term in (141) along the actual comparison path and reporting whether the exponential or metric-compatible branch is active.

B Deferred proofs

This appendix proves the auxiliary identities, inequalities, and functional-analytic facts used in the main text. Each is elementary or classical, but several are load bearing. In particular Lemmas B.4, B.12, and B.13 are the entire route from the training risk to *vocabulary-independent* Levi–Civita transport, so they are proved here rather than asserted.

B.1 The probability-coordinate connection identity

Lemma B.1 (Lower-index α -coefficients). *Let $p : U \rightarrow \Delta_N^\circ$ be C^2 with g_p nondegenerate. Then, in the coordinates of (7)–(8),*

$$\Gamma_{ij,k}^{\text{LC}} = \sum_a \frac{\partial_{ij} p_a \partial_k p_a}{p_a} - \frac{1}{2} C_{ijk}, \quad (144)$$

and consequently (41) holds for every $\alpha \in \mathbb{R}$.

Proof. Differentiating (7),

$$\partial_k g_{ij} = \sum_a \left[\frac{\partial_{ik} p_a \partial_j p_a}{p_a} + \frac{\partial_i p_a \partial_{jk} p_a}{p_a} - \frac{\partial_i p_a \partial_j p_a \partial_k p_a}{p_a^2} \right], \quad (145)$$

and cyclically for $\partial_i g_{jk}$ and $\partial_j g_{ik}$. Collect the Koszul combination $\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}$ term by term. The term $\partial_{ij} p_a \partial_k p_a / p_a$ occurs once in each of the first two expansions and not in the third,

contributing a factor two. The term $\partial_{ik}p_a \partial_j p_a / p_a$ occurs in the first with sign $+$ and in the third with sign $-$, and cancels; likewise $\partial_i p_a \partial_{jk} p_a / p_a$ cancels between the second and the third. The cubic terms contribute $(-1) + (-1) - (-1) = -1$ times C_{ijk} . Hence

$$\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij} = 2 \sum_a \frac{\partial_{ij} p_a \partial_k p_a}{p_a} - C_{ijk}, \quad (146)$$

and halving gives (144). Lowering the final index of (10) with g_p gives $\Gamma_{ij,k}^{(\alpha)} = \Gamma_{ij,k}^{\text{LC}} - \frac{\alpha}{2} C_{ijk}$, which together with (144) is (41). $\square$

Lemma B.2 (Duality of $\pm\alpha$). *For every $\alpha \in \mathbb{R}$, $\partial_k g_{ij} = \Gamma_{ki,j}^{(\alpha)} + \Gamma_{kj,i}^{(-\alpha)}$.*

Proof. By (41) and total symmetry of C ,

$$\begin{aligned} \Gamma_{ki,j}^{(\alpha)} + \Gamma_{kj,i}^{(-\alpha)} &= \sum_a \frac{\partial_{ki} p_a \partial_j p_a + \partial_{kj} p_a \partial_i p_a}{p_a} \\ &\quad - \left(\frac{1+\alpha}{2} + \frac{1-\alpha}{2} \right) C_{ijk}, \end{aligned} \quad (147)$$

and the right-hand side is exactly (145). $\square$

Lemma B.3 (Identification of $\alpha = \pm 1$). *Let $p_\theta(a) = \exp(\theta \cdot T(a) - \psi(\theta))$ be a regular exponential family in minimal natural coordinates θ , and put $\eta_i = \partial_i \psi$. Then*

$$\Gamma_{ij,k}^{(\alpha)} = \frac{1-\alpha}{2} C_{ijk}, \quad g_{ij} = \partial_{ij} \psi, \quad C_{ijk} = \partial_{ijk} \psi. \quad (148)$$

In particular $\nabla^{(1)}$ has vanishing coefficients in θ , so θ is $\nabla^{(1)}$ -affine and constant coordinate vectors are $\nabla^{(1)}$ -parallel; and η is $\nabla^{(-1)}$ -affine, so $\nabla^{(-1)}$ is the mixture connection.

Proof. Here $\log p_\theta(a) = \theta \cdot T(a) - \psi(\theta)$, so $S_i(a) = T_i(a) - \partial_i \psi$ and $H_{ij}(a) = -\partial_{ij} \psi$ is independent of a . Because $\mathbb{E}_p[S_k] = 0$, the first term of (68) vanishes, which gives the first identity of (148). The remaining two are the standard cumulant identities $\mathbb{E}_p[S_i S_j] = \partial_{ij} \psi$ and $\mathbb{E}_p[S_i S_j S_k] = \partial_{ijk} \psi$ for the cumulant generating function ψ . Setting $\alpha = 1$ gives $\Gamma^{(1)} \equiv 0$, proving the first claim.

For $\alpha = -1$, Lemma B.2 with $\Gamma^{(1)} \equiv 0$ gives $\Gamma_{kj,i}^{(-1)} = \partial_k g_{ij} = \partial_{ijk} \psi$, hence $\Gamma^{(-1)l}_{ij} = g^{lk} \partial_{ijk} \psi$. The coordinate change $\theta \mapsto \eta = \nabla \psi(\theta)$ has Jacobian $\partial \eta_i / \partial \theta^j = g_{ij}$, so $\partial / \partial \eta^a = g^{ai} \partial / \partial \theta^i$ and

$$\nabla_a^{(-1)} \partial_b^\eta = g^{ai} \left[(\partial_i g^{bj}) \partial_j + g^{bj} g^{lk} \partial_{ijk} \psi \partial_l \right]. \quad (149)$$

Differentiating $g^{bj} g_{jq} = \delta_q^b$ gives $\partial_i g^{bj} = -g^{bp} g^{qj} \partial_{ipq} \psi$, so the first summand is $-g^{ai} g^{bp} g^{qj} \partial_{ipq} \psi \partial_j$. Renaming $j \rightarrow p$, $l \rightarrow j$, $k \rightarrow q$ in the second summand turns it into $+g^{ai} g^{bp} g^{jq} \partial_{ipq} \psi \partial_j$. The two cancel, so $\nabla_a^{(-1)} \partial_b^\eta = 0$ and η is $\nabla^{(-1)}$ -affine. $\square$

B.2 Elementary inequalities

Lemma B.4 (Hellinger–Kullback). *For $p, q \in \Delta_N^\circ$, $\sum_a (\sqrt{p_a} - \sqrt{q_a})^2 \leq D_{\text{KL}}(p||q)$.*

Proof. Let $\rho = \sum_a \sqrt{p_a q_a}$ be the Bhattacharyya coefficient, so that $\sum_a (\sqrt{p_a} - \sqrt{q_a})^2 = 2(1 - \rho)$. By concavity of the logarithm and Jensen's inequality under p ,

$$-\frac{1}{2} D_{\text{KL}}(p||q) = \sum_a p_a \log \sqrt{\frac{q_a}{p_a}} \leq \log \sum_a p_a \sqrt{\frac{q_a}{p_a}} = \log \rho, \quad (150)$$

that is, $D_{\text{KL}}(p||q) \geq -2 \log \rho$. Since $\log t \leq t - 1$ for $t > 0$ we have $-2 \log \rho \geq 2(1 - \rho)$, and combining the two displays proves the claim. $\square$

Lemma B.5 (Two-model L^1 bound). *Inequality (26) holds.*

Proof. Pointwise in c , the triangle inequality in L^1 and Pinsker's inequality give $\|q_A - q_B\|_1 \leq \sqrt{2D_{\text{KL}}(P\|q_A)} + \sqrt{2D_{\text{KL}}(P\|q_B)}$. Take expectations over C and apply Jensen's inequality to the concave function $t \mapsto \sqrt{t}$, which replaces $\mathbb{E}_C \sqrt{2D_{\text{KL}}(P\|q_i)}$ by at most $\sqrt{2\mathbb{E}_C D_{\text{KL}}(P\|q_i)} = \sqrt{2\epsilon_i}$. The Jensen step is not removable: without it the bound would involve $\mathbb{E}_C \sqrt{D_{\text{KL}}}$, which is not controlled by the excess risk. $\square$

B.3 Structure of the affine softmax head

Lemma B.6 (Kernel and embedding). *Let $F(h) = \text{softmax}(Wh + b)$ with $W \in \mathbb{R}^{N \times d}$, and let $\mathcal{N} = \{u : Wu \in \text{span}\{\mathbf{1}\}\}$. Then*

1. $u^\top G(h)u = \text{Var}_{p_h}(w_Y^\top u)$ for every u , and hence $\ker G(h) = \mathcal{N}$ for every h ;
2. $F(h) = F(h')$ if and only if $h - h' \in \mathcal{N}$;
3. the induced map $\bar{F} : \mathbb{R}^d / \mathcal{N} \rightarrow \Delta_N^\circ$ is a smooth embedding.

Proof. (i) Expanding (14),

$$u^\top G(h)u = \sum_a p_a (w_a^\top u)^2 - \left(\sum_a p_a w_a^\top u \right)^2 = \text{Var}_{p_h}(w_Y^\top u), \quad (151)$$

which vanishes if and only if $a \mapsto w_a^\top u$ is constant on $\{a : p_a > 0\}$. Since p_h is strictly positive this is exactly $Wu \in \text{span}\{\mathbf{1}\}$.

(ii) The softmax map is injective modulo additive constants on logits, so $F(h) = F(h')$ if and only if $W(h - h') \in \text{span}\{\mathbf{1}\}$.

(iii) Fix a reference outcome a_0 and let $L : \Delta_N^\circ \rightarrow \mathbb{R}^{N-1}$ have components $L(p)_a = \log(p_a/p_{a_0})$ for $a \neq a_0$. Then L is a diffeomorphism, its inverse being the softmax that pins coordinate a_0 to zero. Moreover $L \circ F$ is the affine map

$$h \mapsto ((w_a - w_{a_0})^\top h + (b_a - b_{a_0}))_{a \neq a_0}, \quad (152)$$

whose linear part has kernel exactly $\mathcal{N}$ by (ii). Hence $L \circ \bar{F}$ is an injective affine map of the finite-dimensional vector space $\mathbb{R}^d / \mathcal{N}$ onto a closed affine subspace of $\mathbb{R}^{N-1}$; such a map is a homeomorphism onto its image and an immersion. Composing with the diffeomorphism L^{-1} shows that $\bar{F}$ is an injective immersion that is a homeomorphism onto its image, that is, a smooth embedding. (Injective immersions need not be embeddings in general, which is why the intermediate log-odds chart is used.) $\square$

Remark B.7 (Quotients in the general case). Lemma B.6 is elementary because $\mathcal{N}$ is a fixed linear subspace. For a general smooth $F : M \rightarrow \Delta_N^\circ$ no such statement is available: $M / \sim_F$ carries a smooth structure making the projection a submersion only under an additional hypothesis, for instance that F has constant rank and the fiber relation is a closed embedded submanifold of $M \times M$ (Godement's criterion). All estimates in this paper are local and may be read on a chosen horizontal subbundle on which g_F is nondegenerate; the global quotient is never required.

Lemma B.8 (Pullback nondegeneracy). *If $F : M \rightarrow \Delta_N^\circ$ is a smooth immersion then g_F is a Riemannian metric.*

Proof. Since $\sum_a p_a \equiv 1$ we have $\sum_a \partial_i p_a = 0$, so dF takes values in $T_0 = \{u \in \mathbb{R}^N : \sum_a u_a = 0\}$. On T_0 the form $g^\Delta(u, u) = \sum_a u_a^2 / p_a$ is positive definite. Hence $g_F(u, u) = g^\Delta(dFu, dFu) = 0$ forces $dFu = 0$, and $u = 0$ because F is an immersion. $\square$

B.4 The two quantitative steps in Theorem 6.2

Lemma B.9 (Lipschitz dependence on the regularity class). *Fix $\eta, B_2 > 0$ and set*

$$\mathcal{J} = \left\{ (P, P', P'') : P_a \geq \eta, \sum_a P_a = 1, |P'|, |P''| \leq B_2 \right\}. \quad (153)$$

The map sending (P, P', P'') to the tuple (g, Dg, C) assembled from (7), (8), and (145) is Lipschitz on $\mathcal{J}$ with a constant depending only on m, N, η^{-1}, B_2 .

Proof. $\mathcal{J}$ is an intersection of half-spaces, one hyperplane, and closed norm balls, hence convex and compact. Every component of (g, Dg, C) is a rational function of (P, P', P'') whose only denominators are powers of P_a , so the map is C^∞ on the open set $\{P_a > \eta/2\} \supset \mathcal{J}$ and its differential is bounded on the compact set $\mathcal{J}$ by some $L(m, N, \eta^{-1}, B_2)$. Convexity guarantees that the segment joining any two points of $\mathcal{J}$ stays inside $\mathcal{J}$, so the mean-value inequality may be applied along that segment and yields the Lipschitz bound. Convexity is not decorative here: the mean-value inequality fails on a general domain. $\square$

Lemma B.10 (Duhamel with a cancelling growth factor). *Let $\gamma : [0, 1] \rightarrow K$ be piecewise C^1 of Euclidean length L_γ , let $\Gamma, \tilde{\Gamma}$ be continuous connection coefficients with $\|\Gamma\|_{C^0}, \|\tilde{\Gamma}\|_{C^0} \leq M$, and let $P_\gamma, \tilde{P}_\gamma$ denote the induced transports. Then*

$$\|P_\gamma - \tilde{P}_\gamma\|_{\text{op}} \leq L_\gamma e^{ML_\gamma} \|\Gamma - \tilde{\Gamma}\|_{C^0}. \quad (154)$$

Proof. Write $s(t) = \int_0^t |\dot{\gamma}|$, so $s(0) = 0$ and $s(1) = L_\gamma$. Gronwall's inequality applied to $\dot{V} = -\Gamma[\dot{\gamma}]V$ gives $\|P_{1 \leftarrow t}\|_{\text{op}} \leq \exp(M(L_\gamma - s(t)))$ and $\|\tilde{P}_{t \leftarrow 0}\|_{\text{op}} \leq \exp(Ms(t))$. Duhamel's identity writes $P_\gamma - \tilde{P}_\gamma$ as

$$- \int_0^1 P_{1 \leftarrow t} (\Gamma - \tilde{\Gamma})[\dot{\gamma}(t)] \tilde{P}_{t \leftarrow 0} dt, \quad (155)$$

whence

$$\|P_\gamma - \tilde{P}_\gamma\|_{\text{op}} \leq \int_0^1 e^{M(L_\gamma - s(t))} e^{Ms(t)} \|\Gamma - \tilde{\Gamma}\|_{C^0} |\dot{\gamma}(t)| dt. \quad (156)$$

The two exponentials multiply to e^{ML_γ} , which does not depend on t and therefore leaves the integral; the remaining $\int_0^1 |\dot{\gamma}|$ equals L_γ . Bounding each propagator separately by e^{ML_γ} would give the weaker factor e^{2ML_γ} , so this cancellation is what produces the constant stated in (40). If the two coefficient fields carry different bounds M and $\tilde{M}$, the same computation gives $e^{\max\{M, \tilde{M}\}L_\gamma}$. $\square$

B.5 Hilbert-valued Sobolev estimates with dimension-free constants

Theorem 7.2 claims constants independent of the vocabulary size N . Since ψ_p takes values in ℓ_2^N , that claim rests entirely on Sobolev interpolation and embedding for Hilbert-space-valued functions carrying the *scalar* constants: applying the scalar results componentwise would instead produce a factor $\sqrt{N}$ and destroy the claim. We therefore prove both facts. Throughout, $\mathbb{H}$ is a separable Hilbert space, integrals of $\mathbb{H}$ -valued functions are Bochner integrals, and for $s \geq 0$

$$\|f\|_{H^s(\mathbb{R}^m; \mathbb{H})}^2 = \int_{\mathbb{R}^m} (1 + |\xi|^2)^s \|\hat{f}(\xi)\|_{\mathbb{H}}^2 d\xi. \quad (157)$$

Lemma B.11 (Vector-valued Plancherel). *The Fourier transform extends to a unitary operator of $L^2(\mathbb{R}^m; \mathbb{H})$, with no dependence on $\dim \mathbb{H}$.*

Proof. Fix an orthonormal basis (e_n) of $\mathbb{H}$ and write $f = \sum_n f_n e_n$ with $f_n = \langle f, e_n \rangle_{\mathbb{H}} \in L^2(\mathbb{R}^m)$. The Fourier transform acts coordinatewise, $\langle \hat{f}, e_n \rangle = \widehat{f_n}$, and the scalar Plancherel theorem gives $\|f_n\|_{L^2} = \|\widehat{f_n}\|_{L^2}$. By monotone convergence,

$$\|f\|_{L^2(\mathbb{H})}^2 = \sum_n \|f_n\|_{L^2}^2 = \sum_n \|\widehat{f_n}\|_{L^2}^2 = \|\hat{f}\|_{L^2(\mathbb{H})}^2.$$

□

Lemma B.12 (Interpolation with constant one). *For $0 \leq r \leq s$ and $f \in H^s(\mathbb{R}^m; \mathbb{H})$,*

$$\|f\|_{H^r(\mathbb{R}^m; \mathbb{H})} \leq \|f\|_{L^2(\mathbb{R}^m; \mathbb{H})}^{1-r/s} \|f\|_{H^s(\mathbb{R}^m; \mathbb{H})}^{r/s}. \quad (158)$$

Proof. Apply Hölder's inequality with conjugate exponents s/r and $s/(s-r)$ to the factorization

$$(1 + |\xi|^2)^r \|\hat{f}(\xi)\|_{\mathbb{H}}^2 = \left[(1 + |\xi|^2)^s \|\hat{f}(\xi)\|_{\mathbb{H}}^2 \right]^{r/s} \left[\|\hat{f}(\xi)\|_{\mathbb{H}}^2 \right]^{(s-r)/s}, \quad (159)$$

which gives $\|f\|_{H^r}^2 \leq \|f\|_{H^s}^{2r/s} \|f\|_{L^2}^{2(s-r)/s}$; take square roots. The only property of $\mathbb{H}$ used is that $\xi \mapsto \|\hat{f}(\xi)\|_{\mathbb{H}}$ is a nonnegative scalar function, so the constant is one for every $\mathbb{H}$. □

Lemma B.13 (Embedding with a dimension-free constant). *Let $k \in \mathbb{N}_0$ and $r > k + m/2$. Put*

$$\kappa = (2\pi)^{-m/2} \max_{|\beta| \leq k} \left(\int_{\mathbb{R}^m} \frac{|\xi|^{2|\beta|}}{(1 + |\xi|^2)^r} d\xi \right)^{1/2} < \infty. \quad (160)$$

Then for every separable Hilbert space $\mathbb{H}$ and every $f \in H^r(\mathbb{R}^m; \mathbb{H})$, the derivatives $D^\beta f$ with $|\beta| \leq k$ have continuous bounded representatives and

$$\max_{|\beta| \leq k} \sup_{x \in \mathbb{R}^m} \|D^\beta f(x)\|_{\mathbb{H}} \leq \kappa \|f\|_{H^r(\mathbb{R}^m; \mathbb{H})}. \quad (161)$$

Proof. Let first f be an $\mathbb{H}$ -valued Schwartz function. Fourier inversion gives the Bochner integral $D^\beta f(x) = (2\pi)^{-m/2} \int (i\xi)^\beta \hat{f}(\xi) e^{ix \cdot \xi} d\xi$. The triangle inequality for Bochner integrals together with $|\xi^\beta| \leq |\xi|^{|\beta|}$ gives

$$\|D^\beta f(x)\|_{\mathbb{H}} \leq (2\pi)^{-m/2} \int_{\mathbb{R}^m} \frac{|\xi|^{|\beta|}}{(1 + |\xi|^2)^{r/2}} \cdot (1 + |\xi|^2)^{r/2} \|\hat{f}(\xi)\|_{\mathbb{H}} d\xi. \quad (162)$$

Both factors under the integral are nonnegative *scalar* functions of ξ , so the Cauchy–Schwarz inequality in $L^2(d\xi)$ applies and yields $\|D^\beta f(x)\|_{\mathbb{H}} \leq \kappa_\beta \|f\|_{H^r(\mathbb{R}^m; \mathbb{H})}$, where κ_β is the corresponding entry of (160). In polar coordinates the defining integral converges exactly when $2|\beta| - 2r + m < 0$, that is when $r > |\beta| + m/2$, which holds for every $|\beta| \leq k$ by hypothesis. The bound is uniform in x and the integrand is dominated independently of x , so dominated convergence makes each $D^\beta f$ continuous. Since $\mathbb{H}$ -valued Schwartz functions are dense in $H^r(\mathbb{R}^m; \mathbb{H})$ and the estimate is continuous in the H^r norm, it extends to all of $H^r(\mathbb{R}^m; \mathbb{H})$, the limit supplying the continuous representative.

No step refers to $\dim \mathbb{H}$. The Hilbert structure enters only through the Bochner triangle inequality and Lemma B.11, while κ is an integral of a scalar function over $\mathbb{R}^m$. A componentwise argument, by contrast, would apply the scalar embedding to each of the N coordinates and sum, producing an unavoidable factor $\sqrt{N}$. □

Lemma B.14 (Transfer to domains and compact manifolds). *Let $\Omega \subset \mathbb{R}^m$ be a bounded Lipschitz domain, or more generally a Sobolev-extension domain for the range of orders used. Then Lemmas B.12 and B.13 hold on Ω with constants $C(\Omega, m, k, r, s)$ that are again independent of $\mathbb{H}$. The same holds on a compact m -dimensional Riemannian manifold, with constants additionally depending on a fixed finite atlas and the background metric.*

Proof. Stein's extension operator E is a *total* extension operator: it is bounded $H^j(\Omega) \rightarrow H^j(\mathbb{R}^m)$ simultaneously for all $0 \leq j \leq s$, with norms C_j , and it is built from a fixed scalar integral kernel together with reflections. It therefore acts on $\mathbb{H}$ -valued functions by the same formula, commutes with $\langle \cdot, e_n \rangle_{\mathbb{H}}$, and has the same norms C_j for every H . For interpolation,

$$\begin{aligned} \|f\|_{H^r(\Omega; \mathbb{H})} &\leq \|Ef\|_{H^r(\mathbb{R}^m; \mathbb{H})} \leq \|Ef\|_{L^2}^{1-r/s} \|Ef\|_{H^s}^{r/s} \\ &\leq C_0^{1-r/s} C_s^{r/s} \|f\|_{L^2(\Omega; \mathbb{H})}^{1-r/s} \|f\|_{H^s(\Omega; \mathbb{H})}^{r/s}, \end{aligned} \tag{163}$$

and for the embedding $\|f\|_{C^k(\bar{\Omega}; \mathbb{H})} \leq \|Ef\|_{C^k(\mathbb{R}^m; \mathbb{H})} \leq \kappa C_r \|f\|_{H^r(\Omega; \mathbb{H})}$. On a compact manifold, take a finite atlas with a subordinate smooth partition of unity, apply the above in each chart to the localized pushforwards, and sum; the transition maps and cut-off functions act as scalar multipliers and contribute constants depending on the atlas alone. $\square$

Remark B.15 (Where vocabulary-independence comes from). In Theorem 7.2 the target is $H = \ell_2^N$. Lemma B.4 converts the chart-integrated risk into an $L^2(\Omega; \ell_2^N)$ bound with the absolute constant 4; Lemmas B.12–B.14 convert that into a $C^2(\Omega; \ell_2^N)$ bound with constants depending only on Ω, m, r, s ; and Theorem 6.5 consumes only $\delta_1, \delta_2, B_1, B_2, \lambda$. No step introduces a factor of N or of $\min_a p_a$. What survives is the dependence on the uniform envelope B and the spectral floor λ , whose deterioration with model scale is the subject of Appendix A.

B.6 Chart transfer for the probability-coordinate estimates

Lemma B.16 (Manifold form of Theorem 6.2). *Let Ω be a compact m -dimensional Riemannian manifold with a fixed finite atlas whose transition maps have uniformly bounded derivatives up to order three, and suppose the coordinate representations of p and $\tilde{p}$ satisfy Assumption 6.1 in every chart with common constants. Then the conclusions of Theorem 6.2 hold on Ω , with C_Γ multiplied by a factor depending only on the atlas, and with Euclidean path length replaced by background length.*

Proof. Every quantity in (38) is tensorial, and a C^k norm computed in one chart is equivalent to the norm computed in another with constants controlled by the transition derivatives; taking the maximum over the finitely many charts gives globally equivalent norms. Applying Theorem 6.2 in each chart and taking the maximum proves the connection estimate. For the transport estimate, subdivide γ into finitely many subarcs each contained in a single chart. On each subarc the Euclidean and background lengths are comparable with atlas constants, and the transports compose, so Lemma B.10 applies to the concatenation. $\square$

B.7 The conditional-expectation projection

Lemma B.17 (Orthogonal projection in the direct integral). *In the setting of Theorem 8.1, let $\mathcal{L}$ be the space of measurable sections U with $U \in T_{F(H)}P$ almost surely and $\mathbb{E}\|U\|_{g_{F(H)}}^2 < \infty$, equipped with $\langle U, V \rangle = \mathbb{E}[g_{F(H)}(U, V)]$, and let $\mathcal{L}_F \subset \mathcal{L}$ be the subspace of sections of the form $Y \circ F(H)$ with Y measurable. Then $\mathbb{E}[\cdot \mid \sigma(F)]$ maps $\mathcal{L}$ into $\mathcal{L}_F$ and is the orthogonal projection onto it; in particular $\mathcal{L}_F$ is closed.*

Proof. Conditional expectations are taken fiberwise in the ambient zero-sum space T_0 , which is legitimate because conditioning on $F(H) = p$ fixes both $T_p P$ and g_p . Conditionally on $F(H) = p$, the section takes values in the closed linear subspace $T_p P \subset T_0$, and a conditional expectation of a random vector supported in a closed convex set lies in that set almost surely; hence $\mathbb{E}[U \mid \sigma(F)] \in T_{F(H)} P$ and belongs to $\mathcal{L}_F$. Idempotence is the tower property. For self-adjointness, observe that $g_{F(H)}$ is $\sigma(F)$ -measurable, so for every $V \in \mathcal{L}_F$

$$\begin{aligned} \langle \mathbb{E}[U \mid \sigma(F)], V \rangle &= \mathbb{E}[g_{F(H)}(\mathbb{E}[U \mid \sigma(F)], V)] \\ &= \mathbb{E}[\mathbb{E}[g_{F(H)}(U, V) \mid \sigma(F)]] = \langle U, V \rangle, \end{aligned} \tag{164}$$

because both $g_{F(H)}$ and V may be moved inside the conditional expectation. A bounded idempotent map whose range satisfies $\langle \Pi U, V \rangle = \langle U, V \rangle$ for all V in that range is the orthogonal projection onto it, and the range of an orthogonal projection is closed. $\square$

Code and manuscript availability

The manuscript source, model-free predictive-geometry implementation, CPU tests, and experimental protocols are available at

<https://github.com/guybass/probalistic-latent-spaces>.

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